

Variance

The variance is given by the square of the deviations from the mean.

The formula for the variance is given as:

$$Variance(S^2) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Example 2 Calculate the variance of the given vector in Example 1.

We can solve this by hand in R as follows:

```
# get the deviations from the mean

weight <- c(71,69,64,65,73,69,68)

# mean and difference

mean_weight <- mean(weight)

# squared differences

diff_mean <- weight-mean_weight

sqd_diff <- diff_mean^2

var_weight <- sum(sqd_diff)/(length(weight)-1)
var_weight
```

```
[1] 9.952381
```

We can use in build R function as follows:

```
weight <- c(71,69,64,65,73,69,68)
var(weight)
```

```
[1] 9.952381
```

Standard Deviation

The standard deviation is the square root of the variance. It returns the variance to its original scale of the variable.

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

In the current example

$$S = \sqrt{9.952381} = 3.154$$

In R we use sd function

```
weight <- c(71,69,64,65,73,69,68)
sd(weight)
```

```
[1] 3.154739
```

Kurtosis

Kurtosis is a statistical measure that describes the distribution of data in terms of its tails or outliers compared to a normal distribution.

Kurtosis values can be categorized as follows:

High kurtosis (Leptokurtic)

- A high kurtosis value indicates a distribution with a peak that is more pronounced and sharper than a normal distribution.
- It implies that the data is more concentrated around the mean.
- This can result in a distribution that appears more peaked and less spread out.

Low Kurtosis (Platykurtic)

- A low kurtosis value indicates a distribution with a flatter peak compared to a normal distribution.
- It suggests that the data is more spread out.
- This can result in a distribution that looks flatter and less peaked than a normal distribution.

$$Kurtosis = \frac{n \times \sum_{i=1}^n (Y_i - \bar{Y})^4}{\left(\sum_{i=1}^n (Y_i - \bar{Y})^2 \right)^2}$$

Where:

- Y_i is the i^{th} variable of the distribution
- $\bar{Y}$ is the mean of the distribution
- n number of observations

Example

Use the following data to calculate the kurtosis value and give an interpretation

26	12	16	56	112	24
----	----	----	----	-----	----

```
# reading in the observation

data <- c(26,12,16,56,112,24)

# difference of the values and the mean

num <- data-mean(data)

kurt <- (length(data) * (sum(num^4)))/((sum(num^2))^2)

kurt
```

[1] 3.050521

Skewness

Skewness is a measure of the asymmetry in the distribution of data.

A skewness value of 0 indicates that the data is perfectly symmetrical with a balanced distribution of values on both sides of the mean (or median).

In a symmetric distribution:

- The mean, median, and mode are all equal
- The left and right tails of the distribution have similar lengths
- The data is evenly spread around the central point forming a bell shaped curve

The formula for the skewness is:

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{\frac{3}{2}}}$$

By setting the default skewness value to 0, it provides a convenient reference point for understanding the degree of asymmetry in a dataset:

1. If the skewness value is greater than 0, it indicates a **right-skewed (positively skewed)** distribution, where the tail on the right side (positive side) of the central point is longer or more pronounced.
2. If the skewness value is less than 0, it indicates a **left-skewed (negatively skewed)** distribution, where the tail on the left side (negative side) of the central point is longer or more pronounced.
3. If the skewness value is exactly 0, it indicates a **symmetric distribution**, as there is no skew in either direction.

Example

Calculate the skewness for the data in the previous example.

```
# reading in the observation

data <- c(26,12,16,56,112,24)

# difference of the values and the mean

num <- data-mean(data)

skew <- ((length(data)) * (sum(num^3)))/((length(data)-1)*(sum(num^2)))^(3/2)

skew
```

```
[1] 0.2723146
```

Data Frames in R

Introduction

A data frame is a table or a two-dimensional array-like structure in which each column contains values of one variable and each row contains one set of values from each column.

The characteristics of a data frame are:

- The column names should be non-empty
- The row names should be unique
- The data stored in a data frame can be numeric, factor or character
- Each column should contain same number of data items

Create a data frame in R

In R we use the function `data.frame()` function to create a Data Frame.

Example Create the following data frame in R

Name	Age	Gender
Juan	22	Male
John	15	Male
Martha	19	Female

Solution

We create the data frame as follows:

```
# reading in the vectors

Name <- c("Juan", "John", "Martha")

Age <- c(22,15,19)

Gender <- c("Male", "Male", "Female")
```

```
Student_data <- data.frame(Name, Age, Gender)
```

```
Student_data
```

```
      Name Age Gender
1   Juan  22   Male
2   John  15   Male
3 Martha  19 Female
```

Access Data Frame Columns

There are different ways to extract columns from a data frame.

We can use `[ ]`, `[[ ]]`, `$` to access specific columns of a data frame in R.

```
# access elements in a data frame
```

```
# extract the first column
```

```
Student_data$Name
```

```
[1] "Juan" "John" "Martha"
```

```
Student_data[1,] # returns a data frame
```

```
      Name Age Gender
1 Juan  22   Male
```

```
Student_data[["Name"]]
```

```
[1] "Juan" "John" "Martha"
```

We can get the measures of central tendency, measures of dispersion of a variable in a data frame as:

```
mean(Student_data$Age) # mean age of students
```

```
[1] 18.66667
```

Combine Data Frames

In R we can use `rbind()` and `cbind()` function to combine two data frames together.

- `rbind()` - combines two data frames vertically
- `cbind()` - combines two data frames horizontally

```
# create a data frame
dataframe1 <- data.frame (
  Name = c("Juan", "Alcaraz"),
  Age = c(22, 15)
)

# create another data frame
dataframe2 <- data.frame (
  Name = c("Yiruma", "Bach"),
  Age = c(46, 89)
)

# combine two data frames vertically

updated <- rbind(dataframe1, dataframe2)

updated
```

	Name	Age
1	Juan	22
2	Alcaraz	15
3	Yiruma	46
4	Bach	89

The cbind() function adds the columns as shown here

```
# create a data frame

dataframe1 <- data.frame (
  Name = c("Juan", "Alcaraz"),
  Age = c(22, 15)
)

# create another data frame

dataframe2 <- data.frame (
  Hobby = c("Tennis", "Piano")
)

# combine two data frames horizontally

updated <- cbind(dataframe1, dataframe2)
print(updated)
```

	Name	Age	Hobby
1	Juan	22	Tennis
2	Alcaraz	15	Piano

Update observations of a data frame

Assume we have made an error in entry of variable we can easily update the respective row as follows:

```
dataframe1 <- data.frame (
  Name = c("Juan", "Alcaraz"),
  Age = c(22, 15)
)

dataframe1[2,2] <- "19"

dataframe1
```

```
      Name Age
1     Juan  22
2 Alcaraz  19
```

Matrices in R

Matrices are two-dimensional data structures in R and are arranged in a rectangular layout. Matrices can contain only one data type.

To create a matrix in R you use the function `matrix` and specify the number of rows and columns

Example of Matrix in R

Create the following matrix in R

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

```
A <- matrix(c(1,2,3,4), ncol = 2, nrow=2, byrow = TRUE)
A
```

```
      [,1] [,2]
[1,]    1    2
[2,]    3    4
```

In the function `byrow = TRUE` R arranges the matrices according to the rows otherwise by columns.

The difference in the output is given here:

```
A <- matrix(c(1,2,3,4), ncol = 2, nrow=2)
A
```

```
      [,1] [,2]
[1,]    1    3
[2,]    2    4
```

Determinant of Matrices

We can obtain the determinant of a matrix using the function `det()` in R